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Relationships Between Muscle Fibre Conduction Velocity and Frequency Parameters of Surface EMG During Sustained Contraction

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Summary. A surface electrode array has been used to investigate the relationship between muscle fibre conduction velocity and the frequency spectrum during sustained isometric contractions of the biceps brachii. Measurement of muscle fibre conduction velocity was made directly, using the zero-crossing time delay method with two pairs of bipolar electrodes. It was found that the average conduction velocity during an intense (12 kg) sustained contraction decreased by about 20% at the end of the contracting period. Except for peak frequency, changes in the spectral parameters decreased in a similar manner. These results indicate that, during fatiguing contraction, spectral modifications are partly due to reduction in the action potential conduction velocity along the muscle fibres.

Key words: Surface EMG – Conduction velocity – Spectral analysis – Electrode array – Sustained contraction

Introduction

A surface EMG is the summation of temporally and spatially dispersed action potentials travelling along the muscle fibres in the vicinity of the electrodes. Action potentials in the muscle fibres of different motor units are normally asynchronous, so a surface EMG has many of the characteristics of a random signal. Its statistical properties are dependent on the number of active motor units and their rate of activation. Therefore, EMG signals recorded from surface electrodes have not been used in clinical situations because of the difficulty in measuring individual action potential parameters such as amplitude, duration, shape or configuration, and pattern of firing.

Numerous attempts have been made to quantify and derive information about motor unit activities from a surface EMG (Person and Libkind 1970;

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Agarwal and Gottlieb 1975). Lindström et al. (1970) have developed a mathematical model for the power density spectrum of a surface EMG which depends on the conduction velocity of the action potential along muscle fibres. Miyano and Sadoyama (1979) and Sadoyama and Miyano (1981) constructed a stochastic model of the surface EMG and derived a standard (measure H) employing the relative mean conduction velocity from spectral analysis. However, whether the shifts of the EMG spectrum during fatigue are mainly due to conduction velocity changes has yet to be established, because of the lack of a method which measures the absolute value of muscle fibre action potential conduction velocity.

Several investigators have attempted to develop a cross correlation method to estimate the time shift between the EMG signals of two pairs of surface electrodes (Sameshima et al. 1977; Nishizono et al. 1979; Naeije and Zorn 1982). However, these methods require complex computation.

Recently, Lynn (1979) proposed a new approach to the measurement of muscle fibre conduction velocity by using triple surface electrodes. This method is based upon a zero-crossing time delay of two EMG signals and uses a digital filter to eliminate erroneous fluctuation. Masuda et al. (1982) improved Lynn's method with reference to the derivatives of the EMG and achieved temporal accuracy. New possibilities for research on muscle fibre conduction velocity have been opened up by time delay techniques applied in the field of surface electromyography.

In the present work, this new technique is used in a detailed study of surface EMGs of the biceps brachii. The relationship between the changes in conduction velocity and the shifts of frequency parameters of the EMG spectrum during fatiguing contractions are examined.

Methods

EMG Recording

Surface EMG's were recorded from the biceps brachii of two healthy subjects during sustained isometric contraction. The surface electrode configuration is illustrated in Fig. 1. Stainless steel electrodes, each 1 mm wide by 10 mm long, were placed 5 mm apart. The longitudinal array electrode was placed on the muscle belly and oriented perpendicularly to the direction of the muscle fibres. In order to avoid an undefined enlargement of the effective electrode contact area, electrode paste was not used in this experiment. Preparation of the skin was restricted to cleansing with alcohol. EMG signals were amplified by a differential amplifier (60 dB) over a bandwidth of about 5 Hz to 1 KHz.

At the beginning of the experiment, the location of the motor end-plate area was determined. Typical examples of eight simultaneously recorded raw EMGs are shown in Fig. 2. The signal of channel 3 has less amplitude compared to the other channels. This is the motor end-plate area which is located between electrodes 3 and 4 (Fig. 1). EMG recordings for data processing were performed in which electrode pairs 4–5 and 5–6 were used for measuring conduction velocity. The pair 4–10, with an 30-mm electrode separation (most commonly used in surface electromyographs), was used for spectral analysis. The selected electrodes were placed at the same side of the motor end-plate of the muscle.

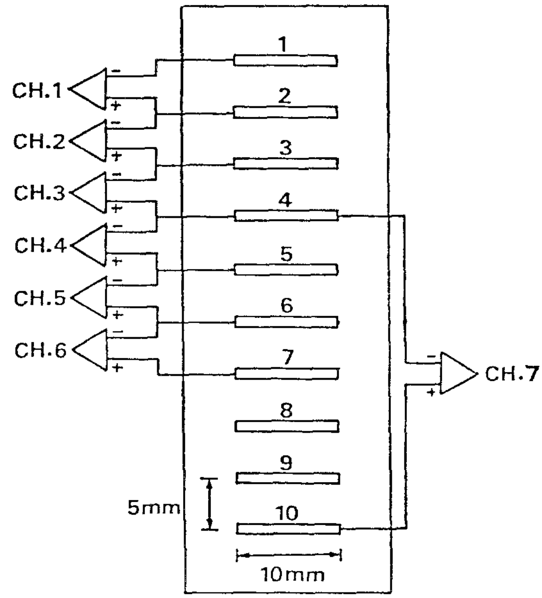

Fig. 1. Schematic diagram of electrode configuration and array

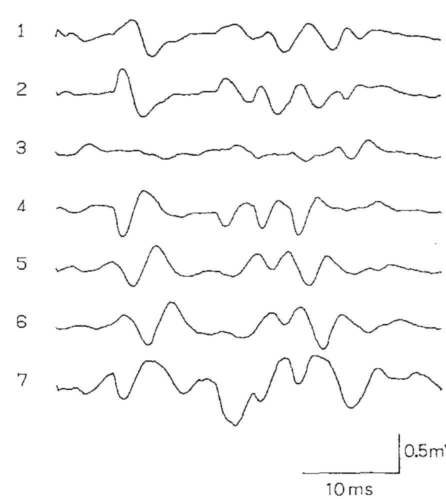

Fig. 2. Typical simultaneous EMGs recorded with a electrode array centered over the motor end-plate area

Processing of EMG Data

Measuring Conduction Velocity. Muscle fibre conduction velocity was measured as described previously (Masuda et al. 1982) by a modified version of Lynn's method (1979). Two bipolar EMG signals were derived from two adjacent electrode pairs. The two signals were similar in form with the time delay (Fig. 3). In calculating the time delay between the two EMG signals, the essential condition was defined as one in which the gradient of the signal is steep at the instant when zero is crossed in the positive direction. The selection was executed with differentiation and threshold circuits. The conduction velocity was calculated from the ratio of electrode spacing (5 mm) to time delay. The actual velocity was estimated as the conduction velocity averaged for 2 s corresponding to the duration of spectral analysis.

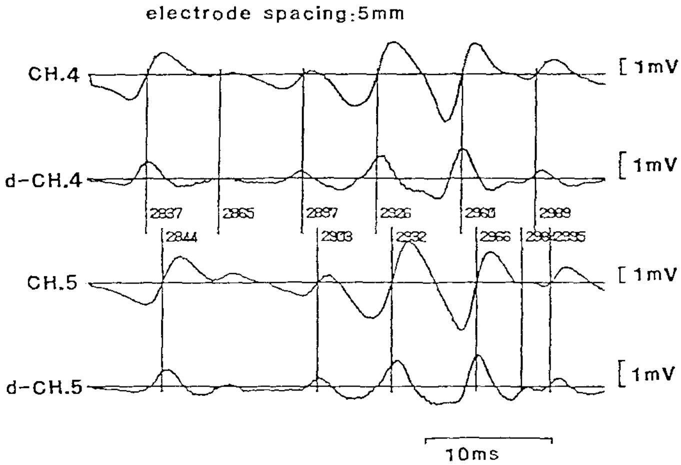

Fig. 3. Zero-crossing time delay measurement with reference to the derivatives of the EMG signals. CHs. 4 and 5 are the raw EMG signals. d-CH.4 and d-CH.5 are the derivatives of CHs.4 and 5, respectively. Vertical lines indicate the occurrences of positive zero-crossings

Spectral Analysis. EMG signals recorded on magnetic tape were digitized at sampling rate of 500 Hz. The time interval between the starting points of successive digitizations was 3 s. The power spectral density function was computed by applying the FFT algorithm. Each spectrum was calculated from a signal period of 2.048 s duration and consisted of 512 points with resolution of 500/1,024 Hz. The spectral parameter were calculated as follows; mean frequency (Petrofsky and Lind 1980; Naeije and Zorn 1982), median frequency (Stulen and DeLuca 1981), measure H (Sadoyama and Miyano 1981), ratio parameter (Stulen and DeLuca 1981), and low frequency ratio below 60 Hz (Kogi and Hakamada 1962).

Testing

Isometric elbow flexion was performed with a load attached to a steel wire connected to a cuff fastened around the subject's wrist. The subjects maintained a specified contraction level. The fatiguing contraction employed was at 4, 8, and 12 kg (equivalent to 50% of maximum voluntary strength) for period of 4, 2, and 1 min, respectively. The subject monitored the required force level on an oscilloscope.

Results

The continuous decrease in conduction velocity during three different contraction levels is shown in Fig. 4. The estimated average conduction velocity was obtained every second. The initial values were 4.28, 4.21, and 4.05 m/s for muscle loading 12, 8, and 4 kg, respectively. The final velocity values at 1, 2, and 4 min during sustained contraction were 3.21, 3.99, and 3.66 m/s, respectively. The decrease in conduction velocity with a muscle fatiguing contraction was found to be significant at 12 kg loading.

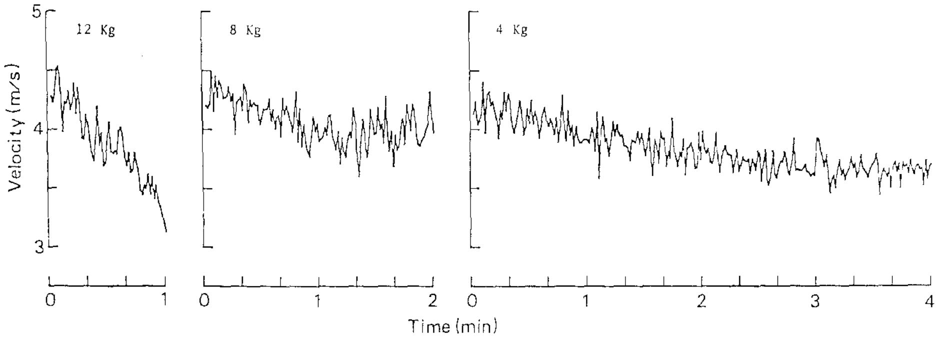

Fig. 4. The changes in conduction velocity as a function of time for different loads (subject A)

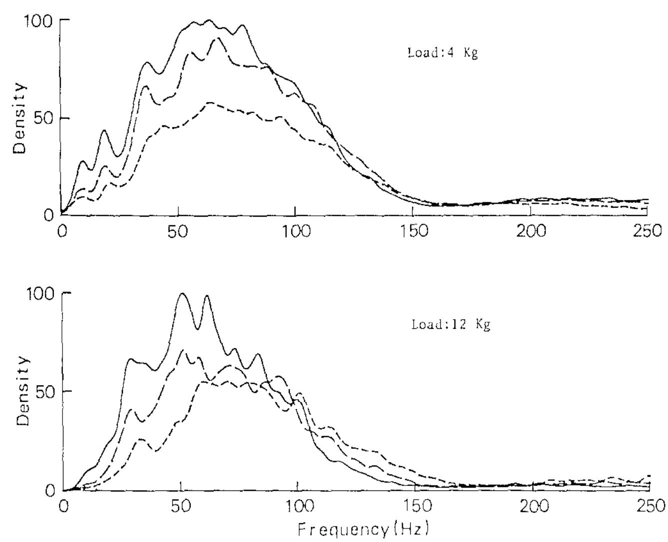

Fig. 5. Example of an EMG power spectrum recorded at the onset (short broken line), the middle (long broken line) and the end (solid line) of three separated session of an isometric fatigue contraction. The power value has been normalized to the peak value of spectral power at all frequencies (subject A)

Figure 5 shows an example of the normalized power spectral density function obtained from bipolar recordings during three different sessions from a sustained contraction at two different contraction levels. With 4 kg loading, the spectra show a power increase with time. With 12 kg loading, significant spectral modifications are shown in the low frequency band. The high frequency region shows a decrease, whereas the low frequency component increases with time. The statistics of the power spectrum with respect to time have been analyzed, as partly shown in Fig. 6. The mean frequency, median frequency, and measure H show significantly faster modification during severe muscle contraction (12 kg)

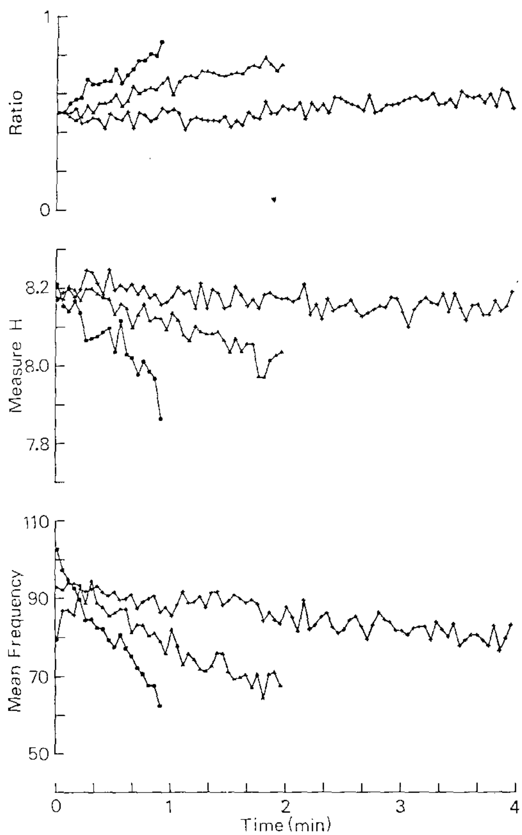

Fig. 6. Changes in spectral parameters (mean frequency, measure H and ratio parameter) for different contraction levels. (subject A) ● : 12 kg, ▲ : 8 kg and + : 4 kg

Table 1. Correlation coefficient between the conduction velocities and the frequency parameters during fatiguing contractions in three different loads for two subjects

Load	4 kg		8 kg		12 kg	
Subject	A	B	A	B	A	B
Mean frequency	0.702	0.578	0.717	0.511	0.947	0.816
Median frequency	0.656	0.456	0.732	0.448	0.922	0.815
Measure H	0.642	0.280	0.589	0.501	0.880	0.732
Peak frequency	− 0.012	0.010	0.361	0.102	0.435	0.597
Low frequency ratio	− 0.645	− 0.407	− 0.712	− 0.456	− 0.936	− 0.840
Ratio parameter	− 0.677	− 0.447	− 0.727	− 0.506	− 0.944	− 0.791

than in the other two conditions. However, the peak frequency fluctuates greatly and is far from being significant. The ratio parameter and the ratio of low frequency components below 60 Hz are found to increase with time.

The relationship between different conduction velocities and frequency parameters during fatiguing contractions with different loads and subjects are summarized in Table 1. Examples of this relationship when the contraction level

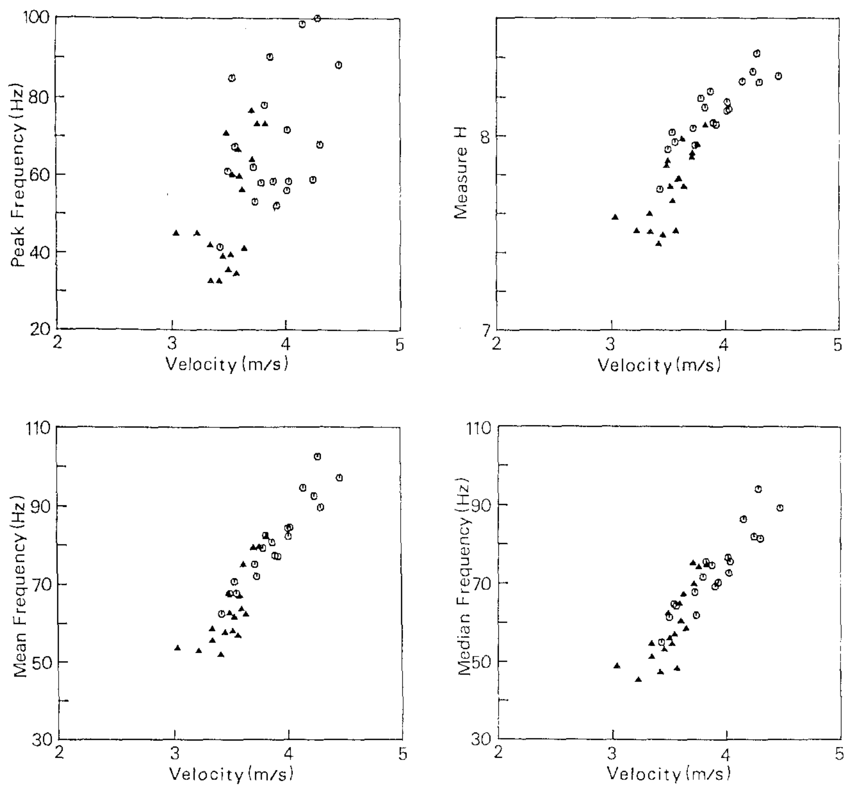

Fig. 7. The relationship between conduction velocity and frequency parameters at 12 kg load. O: subj. A and ▲: subj. B

was at 12 kg loading, which produced the largest modification, are shown in Fig. 7. The relationship was almost linear except for peak frequency. However, the individual difference between the two subjects was apparent in the mean conduction velocity range and frequency parameter range.

Discussion

Muscle action potentials are generated at the motor end-plate and propagate in both directions along the muscle fibre. On the basis of this physiological principle, the present study of muscle fibre conduction velocity was conducted using a surface electrode array. This allowed the simultaneous recording of action potentials from electrodes with well defined mutual separation. Figure 2 clearly shows the propagation of action potentials. In the figure, the signals of channel 3 had less amplitude compared to the other channels. When the motor end-plate lies equidistant from the bipolar electrode poles, the corresponding signal will have a zero value. Furthermore, the signal phases of channels 1 and 2 are symmetrical, as are those of channels 4–6. These signals show time delays, as the location of the recording site is farther from the motor end-plate. This shows that depolarization is generated at the motor end-plate and travels to both fibre ends (Emerson and Zahalak 1981). According to Masuda et al. (1982), the cross correlation coefficient between two adjacent signals away from the motor end-plate was an average of 0.9 or more. This relationship can be recorded by well defined spacing, using electrodes placed perpendicular to the underlying muscle fibre.

The muscle conduction velocities are calculated based upon a zero crossing time delay measurement in two myoelectric signals (Lynn 1979). This method is direct and simple. From a practical point of view, some important questions arise as to how to reject undesired zero crossings caused by noise, and how to determine the processing speed relevant to the electrode spacing. Masuda et al. (1982) solved the problem of faulty detection of zero crossings by using the derivative of the myoelectric signal. When a zero crossing occurs, the derivative of the myoelectric signal is compared with a threshold value, and a zero crossing caused by a small fluctuation is rejected. In addition, these authors showed experimentally that the average conduction velocity is not affected by wide threshold ranges, with the variation of amplitude depending on the contraction levels. Generally, the amplitude increases gradually, even if constant force contractions are maintained. In this case, the threshold level need not be changed for a variety of myoelectric amplitudes, if the electrode spacing is fixed. Temporal accuracy is improved using narrow electrode spacing and a faster clock speed.

A muscle consists of different muscle fibre types with different conduction velocities. Inevitably the conduction velocities obtained every zero crossing are distributed, ranging between 2 and 6 m/s. In this study, estimated conduction velocities were represented as an average for each second. Average velocities at the initial stage, when loaded with 4, 8, and 12 kg, were 3.95, 4.30, and 4.36 m/s, respectively. Slight differences were shown between the contraction level and both subjects. In a contraction where increasing force is required, recruitment of increasingly larger motor units associated with fast fibres will occur (Milner-Brown and Stein 1975). It is suggested that larger motor units associated with faster fibres are recruited with increasing force, and different subjects are different in this respect. The average value of conduction velocity agrees well with previous reports (Buchthal et al. 1955; Nishizono et al. 1979; Lynn 1979; Emerson and Zahalak 1981; Naeiji and Zorn 1982).

Conduction velocity clearly decreased with fatiguing contractions. In the case of the 12 kg load, there is evidence of a reduction of about 20% in velocity at the end of the test. This reduction was somewhat less than the 30% reported by Lynn (1979). Such reductions may be due to the recruitment of new motor units with a slower conduction of action potential (Kadefors et al. 1968; Petrofsky and Lind 1980) and/or the decruitment of active motor units with a faster conduction velocity, due to fatigue. Another hypothesis is a decrease in the propagation velocity of action potentials along the muscle fibres due to the accumulation of lactic acid in the fatigued muscle (Mortimer et al. 1970; Lindström et al. 1977). However further investigation of this matter will be required.

On the other hand, Lindström et al. (1970) showed that conduction velocity can be determined by the electrode spacing and the appearance of a dip in the spectrum. However this method is very sensitive to electrode conditions (Lynn et al. 1978), and it is difficult to detect the dip. For this reason, several investigators have attempted to use the change in frequency parameters as an index of muscle fatigue (Petrofsky and Lind 1980; Viitasalo and Komi 1977; Stulen and DeLuca 1981). The present results indicate that the changes in frequency parameters, mean frequency, median frequency and measure H, ratio parameter and low frequency ratio below 60 Hz were related to changes seen simultaneously in the conduction velocity in severe contractions, but peak frequency was the least reliable among them, as shown by Stulen and DeLuca (1981). In weak contractions, both relationships were less significant. These results indicate that small changes were insufficient to account for both relationships. Although the statistics of a spectrum are indeed partly dependent on the conduction velocity (Bigland-Ritchie et al. 1981), they are also affected by the calculated frequency range and the shape of the spectrum. Furthermore they are also known to vary with the type of electrode used to detect the signal, as well as the placement of the electrode (Lynn et al. 1978; Zipp 1978). Thus, the use of frequency parameters for spectral modifications during fatigue is not suitable from many points of view. In order to obtain information on conduction velocity from the surface EMG spectrum, it is necessary to define the electrode conditions.

Finally the zero crossing time delay method using fewer measuring parameters has provided useful information about the relationships occuring between muscle conduction velocity and muscle fatigue.

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